Muhammad Hasnain Pirzada

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Full Stack AI Engineer — LLM Systems, RAG & Production AI Agents

Full stack AI engineer focused on building production LLM systems: agent orchestration, RAG pipelines, and multi-tenant backends that serve real users, not demos. Recent work includes the backend for a multi-tenant AI platform serving ~200 client businesses (3 apps, 1,400+ API endpoints, 50+ integrations), **6 MCP servers (17 MCPs, 1,000+ tools)** connecting AI agents to production data, and RAG-backed chatbots grounded in each client's own data. Comfortable across the stack: **FastAPI, PostgreSQL, Redis, Claude/OpenAI/Gemini APIs, LangChain, React/Next.js, Docker, AWS**. Currently deepening hands-on work with open-source LLMs — fine-tuning (LoRA/QLoRA) and quantization (GGUF/GPTQ) — to extend into self-hosted, cost-efficient model deployment.

WORK EXPERIENCE

Associate Software Engineer (AI/LLM) — Techtimize *Nov 2025 – Present*

- Built and maintained **40+ FastAPI modules** (website/landing page builder, blog engine, keyword research, content matrix, chatbot, appointments, social publishing) powering a multi-tenant AI platform used by **~200 client businesses**.
- Built and deployed **6 production Model Context Protocol (MCP) servers (17 MCPs, 1,000+ tools)** integrating AWS S3, PostgreSQL/MySQL, WhatsApp Business, Salesforce, and WordPress, replacing manual data operations with agent-driven workflows.
- Built AI chatbot systems on **Claude API, LangChain, and pgvector-backed RAG**, answering from each client's own documents with context-aware retrieval and automated response generation.
- Built voice AI pipelines combining **Whisper, Deepgram, ElevenLabs, and Retell AI** for real-time conversational automation, replacing manual client call handling.
- Built a **Redis-backed job layer** sharing API rate limits across processes and parallelizing keyword research jobs, cutting one regional research step from **~9 minutes to ~60 seconds** (a ~9× speedup).
- Found that AI/API cost tracking was overstating real spend by **~15×**; rebuilt it as a single cost-tracking system on one unified pricing table across all models, giving the team accurate per-client cost data for the first time.

Associate Software Engineer — Cognisons *Sept 2025 – Nov 2025*

- Worked end-to-end on ML pipelines — feature engineering, model training/validation (**PyTorch, TensorFlow, Scikit-learn**), and model explainability (**SHAP**) — for production NLP and LLM-powered features.
- Built scalable web applications using **React and FastAPI** for real-world business use cases.

Internship — Applied AI & NLP, ITSOLERA

- Built a conversational agent for automated query response, fine-tuned a BERT model for question answering, and implemented real-time AI-powered language translation.

FEATURED PROJECT

AI Marketing & Operations Platform for a US Digital Agency (Home-Service Businesses) | *Python, FastAPI, NestJS, React, PostgreSQL, Redis, Claude/OpenAI/Gemini*

Multi-tenant platform a digital agency uses to run the entire online presence of ~200 local home-service businesses (roofers, plumbers, HVAC). Client data lives in one shared source of truth; AI (Claude, GPT, Gemini) drafts websites, blogs, SEO/keyword plans, and Google Business Profile content from it, publishing through a human-approval step at every stage. The platform spans 3 apps and 1,400+ API endpoints: a React dashboard, a Node.js/NestJS backend, a Python/FastAPI AI backend, and an MCP layer exposing platform data to AI agents.

- Built the **AI content generation pipeline** producing website copy, blog posts, and SEO/keyword plans from each client's structured profile data, using Claude, GPT, and Gemini across content types — AI output saves as a draft and a person approves it before publishing, with each generated row recording which prompt version produced it.
- Built a **Google Business Profile console** managing **~200 client listings**, reconciling live Google data against client-approved data across hours, categories, services, attributes, logos, and performance insights.
- Built **MCP tool servers** (part of a 6-server, 1,000+ tool agent-facing layer) giving AI agents structured access to live platform data under one standard response format.

- Worked across **50+ integrations**, including Google Search Console, PageSpeed/CrUX, DataForSEO, Salesforce, WordPress, Housecall Pro, FieldRoutes, Retell AI, Zoom, JotForm, Meta, and Pinterest.
- Built a **project reconciliation engine** auto-matching finished-job photos and records from field crews (CompanyCam, CrewCam, Housecall Pro, FieldRoutes) to the correct client project via an AI-drafted match, cutting manual matching down to a single human approval click.
- Migrated legacy raw SQL onto the **SQLAlchemy 2.0 ORM** and managed schema evolution through versioned **Alembic** migrations.

OTHER PROJECTS

Image Super-Resolution using Vision Transformer (LMLT) | *Final Year Project — PyTorch*
Code: github.com/engrhasnain/SRViT-FYP

- Built an LMLT (Low-to-High Multi-Level Transformer) model for brain MRI super-resolution, reaching **PSNR 46.73 / SSIM 0.9948** at 2× and 4× upscaling on 1,596 grayscale scans, outperforming SRCNN, SRGAN, and SwinIR baselines.

Website Classification — Full-Stack ML App | *FastAPI, OCR, SVM, React*
Code: github.com/engrhasnain/fastapi-ml-predictor

- Built an OCR + TF-IDF pipeline classifying website screenshots (educational/job/e-commerce) via an SVM model at **90% accuracy**, served through FastAPI with a React (Vite) + Tailwind frontend.

TECHNICAL SKILLS

- **Programming:** Python, JavaScript/TypeScript, C++, R, SQL
- **Backend & APIs:** FastAPI, NestJS, Node.js, FastMCP, REST, GraphQL, Salesforce API, DataForSEO API, WordPress API
- **LLMs & AI Tools:** Anthropic Claude, OpenAI GPT, Google Gemini, LangChain, LangGraph, RAG, Agents, Model Context Protocol (MCP), Groq, Hugging Face
- **Open-Source LLMs (applied learning):** hands-on fine-tuning (LoRA/QLoRA) and quantization (GGUF/GPTQ) of open-source models (Llama, Mistral), self-directed
- **Voice AI:** Whisper, Deepgram, ElevenLabs, Retell AI, real-time STT/TTS pipelines
- **Databases & Vector Stores:** PostgreSQL, MySQL, Redis, MongoDB, AWS S3, pgvector, Pinecone, Weaviate
- **Frontend:** Next.js, React, TypeScript, Tailwind CSS, Streamlit
- **ML/DL:** PyTorch, TensorFlow, Scikit-learn, SHAP, Hugging Face Transformers
- **DevOps & Cloud:** Docker, Docker Compose, AWS (EC2, S3), Git, GitHub, CI/CD

EDUCATION

University of Engineering & Technology, Mardan (UET-M) *Mardan, Pakistan*
Bachelor of Engineering, Computer Software Engineering — Sept 2021 – June 2025 | CGPA: **3.76/4.0** | PEC Level-II recognized, OBE degree (PLO score: 80%)

CERTIFICATIONS

- Machine Learning Specialization — Stanford University (3 courses)
- Applied Data Science with Python — University of Michigan (5 courses)
- Mastering REST APIs with FastAPI — Packt / Coursera
- Neural Networks and Deep Learning — DeepLearning.AI
- Complete Generative AI with LangChain and Hugging Face — Udemy

LANGUAGES

English (Fluent) | Urdu (Fluent) | Pashto (Fluent)